

MINIMUM LAP TIME OPTIMIZATION

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ABSTRACT.

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1. THE PROBLEM SETUP

The objective of this project is to compute the minimum-time trajectory for a race car on a closed racing circuit. Unlike conventional path-following problems, where the trajectory is predefined, both the vehicle trajectory and the control inputs are optimized simultaneously while satisfying the physical limitations of the vehicle and the geometric constraints imposed by the track.

The optimization is performed on a real karting circuit located in Freiburg. The track geometry is represented by its centerline, local curvature and track width, which constitute the geometric input to the optimal control problem. These quantities are obtained by processing an image of the circuit and extracting its boundaries.

Besides the track geometry, the relevant vehicle parameters are assumed to be known. These include the wheelbase, steering angle limits, longitudinal acceleration and braking limits, and the maximum admissible lateral acceleration. Together, these parameters define the dynamic and physical constraints of the optimization problem.

The objective is therefore to determine the state and control trajectories that minimize the total lap time while ensuring that the vehicle remains within the track boundaries and satisfies all vehicle constraints.

To formulate this optimal control problem, an appropriate mathematical model of the vehicle is required. In the following section, a simplified kinematic bicycle model is introduced together with the assumptions adopted throughout this work.

2. VEHICLE MODEL

The vehicle model must capture the main relationship between steering, longitudinal motion, and the resulting trajectory while remaining sufficiently simple for the numerical optimization problem. Since this project focuses on the minimum-lap-time optimization problem rather than on detailed vehicle dynamics, we use a kinematic bicycle model.

The kinematic bicycle model describes the vehicle motion using only a few states and control inputs. It is simple enough for numerical optimization while still capturing the main effects of steering and acceleration. More detailed effects, such as tire slip, load transfer, suspension motion, and transient tire forces, are neglected.

2.1. The bicycle model.

The commonly used kinematic bicycle model is a simplification of the dynamics of a car. It represents the two wheels of each axle by a single equivalent wheel, resulting in one front wheel and one rear wheel. Only the front wheel is used for steering, and tire slip is neglected.

The model is described by the position of the car in global coordinates (x, y) , the velocity v , the heading ψ , and the steering angle δ . The dynamics of the model are given by

$$\begin{aligned}\dot{x} &= v \cos(\psi) \\ \dot{y} &= v \sin(\psi) \\ \dot{\psi} &= \frac{v}{L} \tan(\delta) \\ \dot{v} &= a\end{aligned}$$

The parameter L denotes the wheelbase of the car, i.e. the distance between the front and rear wheel, and the acceleration of the car respectively. For more details on this model, see [Raj12].

However, for driving around a given track it is inconvenient to use global coordinates because it is more complicate to formulate the constraints. Additionally, we only need the position on the track and do not care for the absolute position in global coordinates.

Therefore, we want to formulate the model using track coordinates. The centerline distance traveled will be given as s and e_y denotes the distance from the centerline, e_ψ denotes the angle between the cars heading and the tangent of the centerline and v still denotes the velocity of the vehicle.

The position of the vehicle can be expressed relative to the track centerline as

$$\mathbf{p}(t) = \mathbf{p}_c(s(t)) + e_y(t)\mathbf{N}(s(t)),$$

where $\mathbf{p}_c(s)$ denotes the centerline position and $\mathbf{N}(s)$ is its unit normal vector.

Differentiating with respect to time gives

$$\dot{\mathbf{p}} = \frac{d\mathbf{p}_c}{ds}\dot{s} + \dot{e}_y\mathbf{N} + e_y\frac{d\mathbf{N}}{ds}\dot{s}.$$

Using

$$\frac{d\mathbf{p}_c}{ds} = \mathbf{T}, \quad \frac{d\mathbf{N}}{ds} = -\kappa(s)\mathbf{T},$$

where $\mathbf{T}(s)$ is the unit tangent vector, we obtain

$$\dot{\mathbf{p}} = (1 - \kappa(s)e_y)\dot{s}\mathbf{T} + \dot{e}_y\mathbf{N}.$$

The vehicle velocity can also be decomposed into components tangent and normal to the centerline:

$$\dot{\mathbf{p}} = v \cos(e_\psi) \mathbf{T} + v \sin(e_\psi) \mathbf{N}.$$

Equating the tangential and normal components gives

$$\begin{aligned} (1 - \kappa(s)e_y) \dot{s} &= v \cos(e_\psi), \\ \dot{e}_y &= v \sin(e_\psi). \end{aligned}$$

Therefore,

$$\dot{s} = \frac{v \cos(e_\psi)}{1 - \kappa(s)e_y}.$$

This relation is commonly used in Frenet-coordinate formulations of vehicle dynamics [RNFD23].

The heading error is defined as

$$e_\psi = \psi - \psi_c(s),$$

where $\psi_c(s)$ denotes the heading of the track centerline.

Differentiating with respect to time and applying the chain rule yields

$$\dot{e}_\psi = \dot{\psi} - \frac{d\psi_c}{ds} \dot{s}.$$

The cars own heading with respect to the tangent of the centerline changes because of steering and the heading of the track changes because of the curvature of the track.

Since the derivative of the centerline heading with respect to the arc length is equal to the local track curvature,

$$\frac{d\psi_c}{ds} = \kappa(s),$$

the heading-error dynamics become

$$\dot{e}_\psi = \frac{v}{L} \tan(\delta) - \kappa(s) \dot{s}.$$

This results in the following vehicle model expressed in Frenet coordinates. The state vector is defined as

$$\mathbf{z}(t) = \begin{bmatrix} s(t) \\ e_y(t) \\ e_\psi(t) \\ v(t) \end{bmatrix},$$

where $s(t)$ denotes the progress along the track centerline, $e_y(t)$ the lateral offset from the centerline, $e_\psi(t)$ the heading error, and $v(t)$ the vehicle velocity.

The control vector is given by

$$\mathbf{u}(t) = \begin{bmatrix} a(t) \\ \delta(t) \end{bmatrix},$$

where $a(t)$ is the longitudinal acceleration and $\delta(t)$ is the steering angle.

The state dynamics can then be written compactly as

$$\dot{z}(t) = f(z(t), u(t)) = \begin{bmatrix} \frac{v(t) \cos(e_\psi(t))}{1 - \kappa(s(t))e_y(t)} \\ v(t) \sin(e_\psi(t)) \\ \frac{v(t)}{L} \tan(\delta(t)) - \kappa(s(t)) \frac{v(t) \cos(e_\psi(t))}{1 - \kappa(s(t))e_y(t)} \\ a(t) \end{bmatrix}.$$

The kinematic bicycle model neglects tire slip, load transfer, suspension dynamics and transient tire forces. Consequently, it is not intended to accurately describe the vehicle behaviour at the limits of adhesion.

Nevertheless, it captures the main relationship between steering, longitudinal acceleration and the resulting vehicle trajectory while keeping the optimization problem sufficiently simple.

3. OPTIMIZATION PROBLEM

3.1. Formulation of the optimization problem.

Using the vehicle model introduced in the previous section, the minimum-lap-time problem can be formulated as a continuous-time optimal control problem. The decision variables are the state trajectory $z(\cdot)$, the control trajectory $u(\cdot)$, and the final time T .

The control vector

$$u(t) = \begin{bmatrix} a(t) \\ \delta(t) \end{bmatrix}$$

contains the longitudinal acceleration and the steering angle.

The objective is to minimize the total time required to complete the track. The resulting optimal control problem is

$$(3.1) \quad \underset{z(\cdot), u(\cdot), T}{\text{minimize}} \quad T$$

$$(3.2) \quad \text{subject to} \quad \dot{z}(t) = f(z(t), u(t)), \quad t \in [0, T],$$

$$(3.3) \quad -\frac{w(s(t)) - b}{2} \leq e_y(t) \leq \frac{w(s(t)) - b}{2}, \quad t \in [0, T],$$

$$(3.4) \quad -a_{\min} \leq a(t) \leq a_{\max}, \quad t \in [0, T],$$

$$(3.5) \quad -\delta_{\max} \leq \delta(t) \leq \delta_{\max}, \quad t \in [0, T],$$

$$(3.6) \quad 0 \leq v(t) \leq v_{\max}, \quad t \in [0, T],$$

$$(3.7) \quad -a_{y,\max} \leq \frac{v(t)^2}{L} \tan(\delta(t)) \leq a_{y,\max}, \quad t \in [0, T],$$

$$(3.8) \quad 1 - \kappa(s(t))e_y(t) > 0, \quad t \in [0, T],$$

$$(3.9) \quad s(0) = 0, \quad s(T) = S,$$

$$(3.10) \quad v(0) = 0, \quad e_y(0) = 0, \quad e_\psi(0) = 0,$$

$$(3.11) \quad T > 0.$$

The parameters appearing in the optimal control problem are defined as follows:

$w(s)$: track width at centerline position s ,

b : vehicle width,

L : vehicle wheelbase,

$a_{\min} > 0$: magnitude of the maximum permissible deceleration,

$a_{\max} > 0$: maximum permissible acceleration,

$\delta_{\max}$: maximum absolute steering angle,

$v_{\max}$: maximum permissible velocity,

$a_{y,\max}$: maximum permissible lateral acceleration,

S : total centerline length of the track.

The objective function (3.1) minimizes the final time T . The dynamic constraint (3.2) ensures that the state trajectory is consistent with the kinematic bicycle model.

The track constraint (3.3) restricts the center of the vehicle to the usable track width. The vehicle width b is subtracted from the track width so that the complete vehicle, rather than only its reference point, remains inside the track boundaries.

The acceleration, steering, and velocity constraints (3.4)–(3.6) represent the longitudinal and kinematic limitations of the vehicle. The lateral acceleration constraint (3.7) limits the cornering acceleration generated by the steering input.

The boundary conditions require the vehicle to start at the beginning of the track and reach the total centerline distance S . In the present formulation, the vehicle starts from rest at the centerline with zero heading error.

4. NUMERICAL SOLUTION

We need to cover here:

- (1) Discretization
- (2) Multiple-shooting
- (3) CasAdi
- (4) IPOPT (or other solver use)

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